



ESRF STATUS REPORT

Antonia BETEVA

On behalf of the ESRF MXCuBE team:

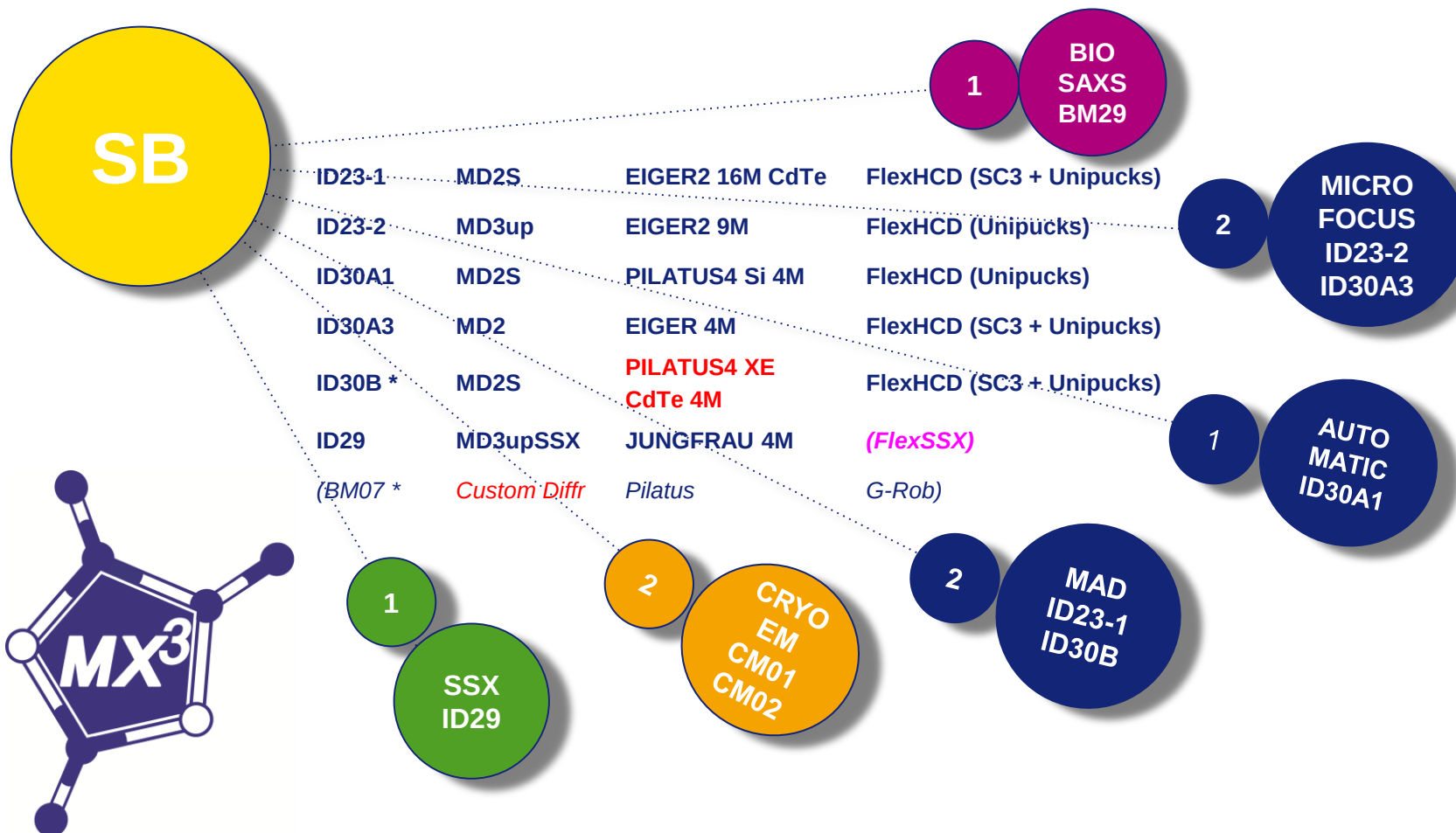
Marcus Oskarsson, Daniele de Sanctis, Yan Walesch, ,
Didier Nurizzo, Mathis Loussert, Antonia Beteva

And contributors:

Olof Svensson, Ludovic Broche, Alex de Maria



BRIEF BEAMLINES SUMMARY



* ID30B New detector Pilatus4 XE CdTe 4M in operation

* BM07 New Diffractometer in production (Aerotech, b-zoom, Wago)



MXCuBE (same on all beamlines)

- mxcubeweb 4.683.0 + ESRF (260421)
- mxcubecore 2.66.0 + ESRF (260505)

Keeping up with the develop branch with a two weeks delay,
At last meeting we were at:

- mxcubeweb 4.551.0
- mxcubecore 1.396.0

Other deployment

- New AbstractDiffractometer – SampleView on all the beamlines
- Python 3.11, conda environment
- Automated data collection on several beamlines



Cybersecurity tools provided by ESRF infrastructure

- Vulnerability Scanner (metrics)
- Web Application Firewall (WAF) – external application
- Network segmentation
- SSO & LDAP

Cybersecurity for MXCuBE (see M. Loussert's presentation)

- Implement cybersecurity best practices in all development levels (development process, source code and deployment)
- Update third party library versions to avoid vulnerabilities.



Done and ongoing

- AbstractDiffractometer – SampleView (core and web).
- MXCuBE-SSX on ID29 – data collection for chip, foils, injector and tape drive.
- Port data collection routines to Bliss - ongoing.
- Replace MicrodiffBeamstop and MicrodiffLightInOut by MicrodiffLightBeamstop
- MASSIF-1 - GPhL Workflow
- Camera hutch view in MXCuBE



Testing, deployment and cybersecurity related

- Improved testing environment using the MD, Flex, Lima simulation Docker images. To be used as well for pentests
- Continue to refactor and modernise front and back end code.
- Move to React 18.
- Cybersecurity audit “executive summary” (see M.Oscarsson’s presentation)



Ongoing

- Spot finding mechanism for Pilatus4 with Lima2
- Installation of camera hutch view in MXCuBE, using Argus on all the missing beamlines.
- ID29 SSX tape speed calculator in Argus, tape drive controller
- Automation Working group - Implementation of the X-Ray Scan.
- UX/UI Working group - Work on multi user automated data collection and overall improvements to the UX



New development

- ID29 SSX features - pump & probe
- UX/UI Working group - Work on multi user automated data collection and overall improvements to the UX
- Pentest – near future